

## **Lecture 9**

### **Correlation, Regression**

Part 1

Models, Association, and Linear Regression

June 2026

## WHAT IS A MODEL?

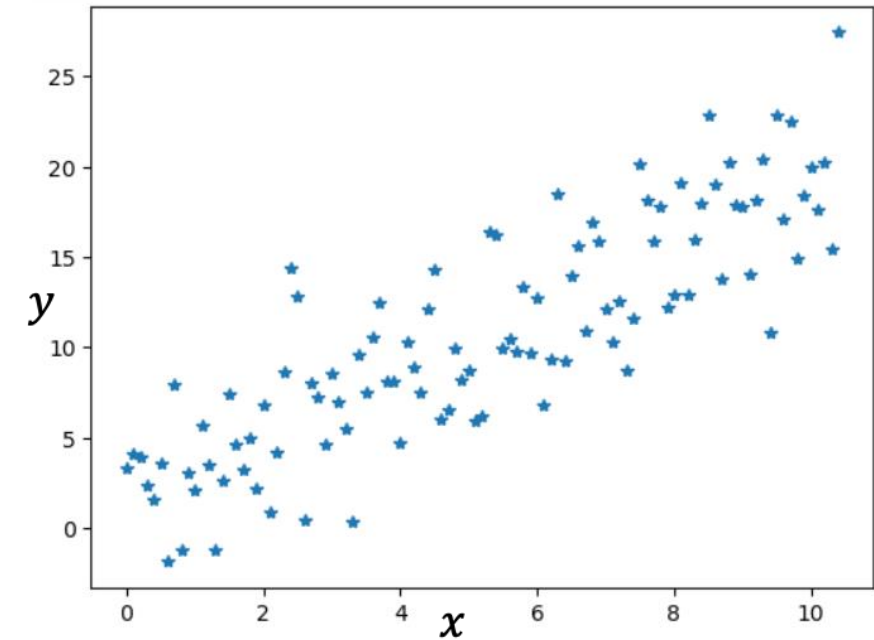
A model is a mathematical representation of a simplified aspect of reality. It allows us to understand, measure, and predict quantities of interest.

Given certain inputs, the model is used to calculate or estimate the output.

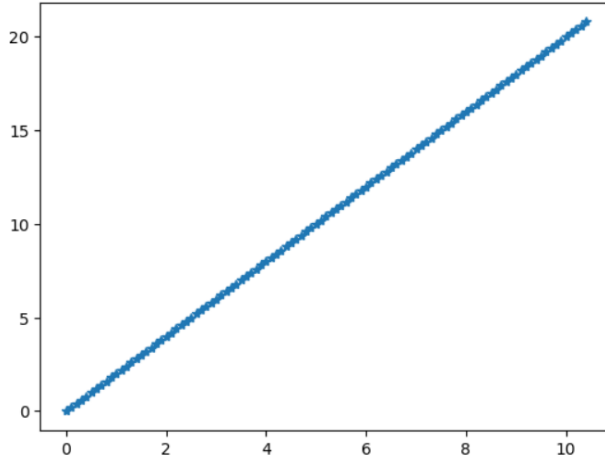
For instance, in this example, the model that generated the points was:

$$y = 2x + (\text{noise})$$

Typically, we have data (inputs and outputs), and we want to find the model that best fits the real data. If the fit is useful, we can use the model as a predictor for new observations.



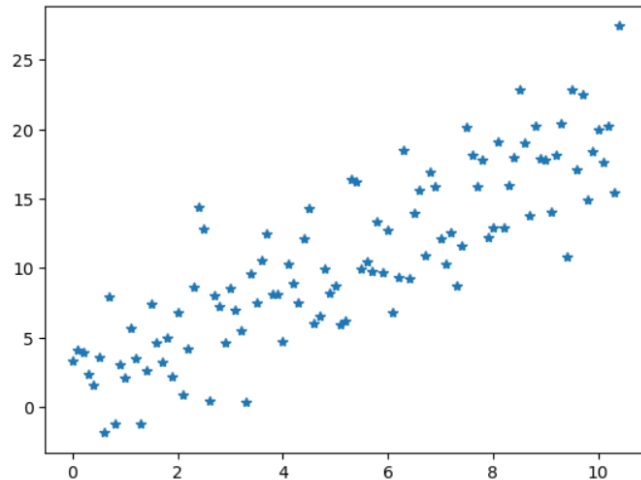
## DETERMINISTIC VS STATISTICAL (NOISY MODEL)



Deterministic model: knowing the inputs, one can calculate exactly the output.  
Deterministic models are easy to use, but they often simplify reality too much.

Statistical model: knowing the inputs, one can estimate a range, or a distribution, in which the output is likely to fall.

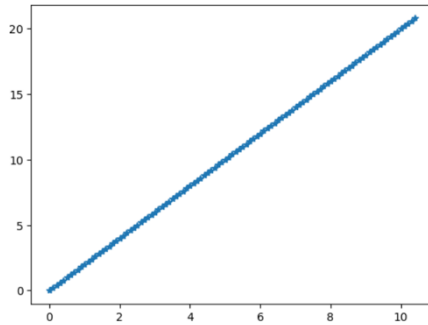
The noise in the model usually captures information that we do not observe: variables that affect the output but are not included in the model.



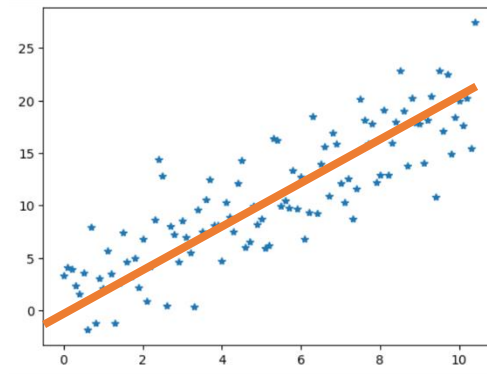
## LINEAR VS NONLINEAR?

Linear model: the predicted average value follows a straight line. When the model is noisy, a linear model means that the mean of the distribution for a given input value lies on a straight line.

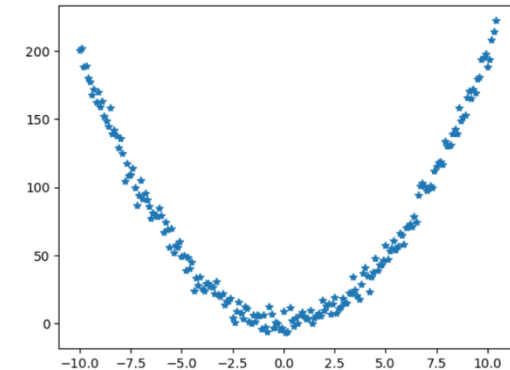
Nonlinear model: any model where the predicted average value does not follow a straight line.



Linear  
deterministic



Linear noisy



Nonlinear noisy

## SIMPLE VS MULTIPLE REGRESSION

Simple regression: the model uses one explanatory variable.

Multiple regression: the model uses several explanatory variables. For instance, we could predict satisfaction using amount, age, items bought, and whether a credit card was used.

Some business models have hundreds of inputs.

| inputs |        |     |       |      | output       |
|--------|--------|-----|-------|------|--------------|
| id     | amount | age | items | card | satisfaction |
| 1      | 163.5  | 43  | 7     | 0    | good         |
| 2      | 138.8  | 39  | 5     | 0    | very good    |
| 3      | 175.9  | 48  | 7     | 1    | low          |
| 4      | 157.5  | 45  | 11    | 1    | low          |
| 5      | 60.0   | 43  | 2     | 0    | very good    |
| ...    | ...    | ... | ...   | ...  | ...          |

id is not used  
in the model

## INPUTS AND OUTPUTS DEPEND ON THE QUESTION

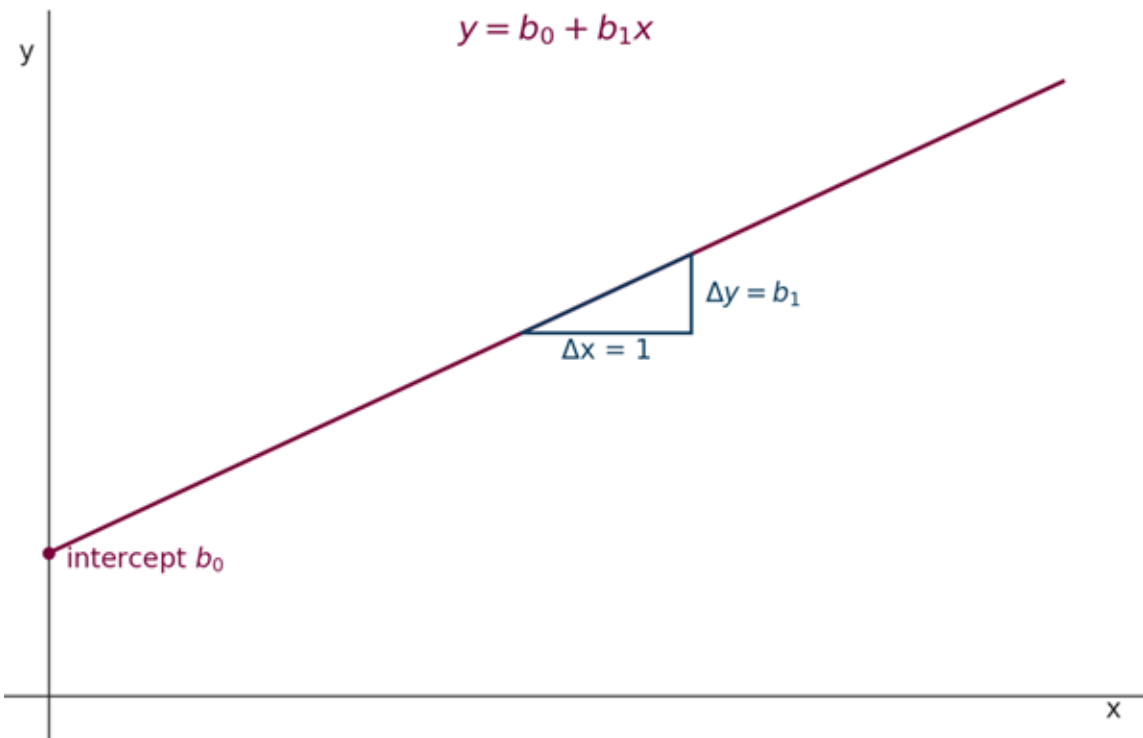
Using the same dataset, we can build different predictive models depending on what we want to predict or understand.

What counts as an input and what counts as the output depends on the objective of the analysis.

|     | output | inputs |       |      |              |
|-----|--------|--------|-------|------|--------------|
| id  | amount | age    | items | card | satisfaction |
| 1   | 163.5  | 43     | 7     | 0    | good         |
| 2   | 138.8  | 39     | 5     | 0    | very good    |
| 3   | 175.9  | 48     | 7     | 1    | low          |
| 4   | 157.5  | 45     | 11    | 1    | low          |
| 5   | 60.0   | 43     | 2     | 0    | very good    |
| ... | ...    | ...    | ...   | ...  | ...          |

not used  
here

## THE LINEAR MODEL



A linear model describes an exact straight-line relationship between  $x$  and  $y$

$$y = b_0 + b_1x$$

Here :

$b_0$  = intercept

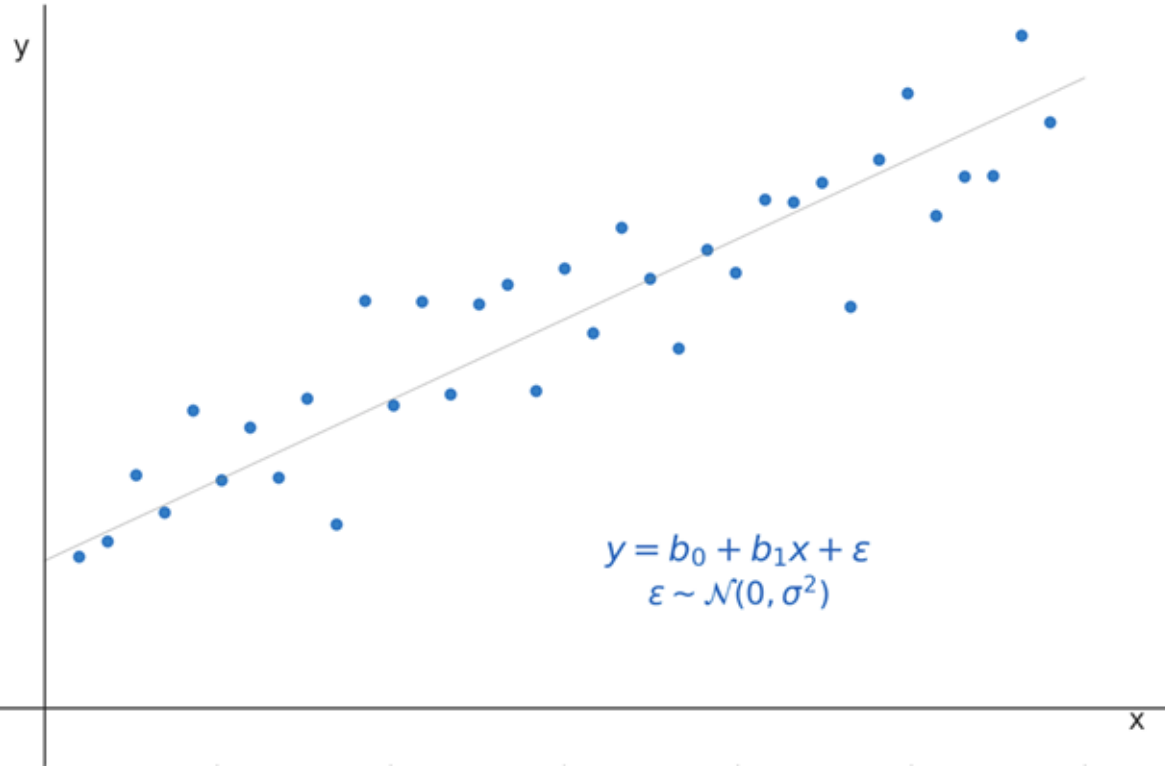
$b_1$  = slope

This is a deterministic line: once  $x$  is known,  $y$  is determined exactly.

*Sometimes the linear model is written using the beta notation (Greek notation) as*

$$y = \beta_0 + \beta_1x$$

## STATISTICAL LINEAR MODEL



A statistical model keeps the linear part, but adds a noise term for the unexplained variation:

$$y_i = b_0 + b_1x_i + \varepsilon_i$$

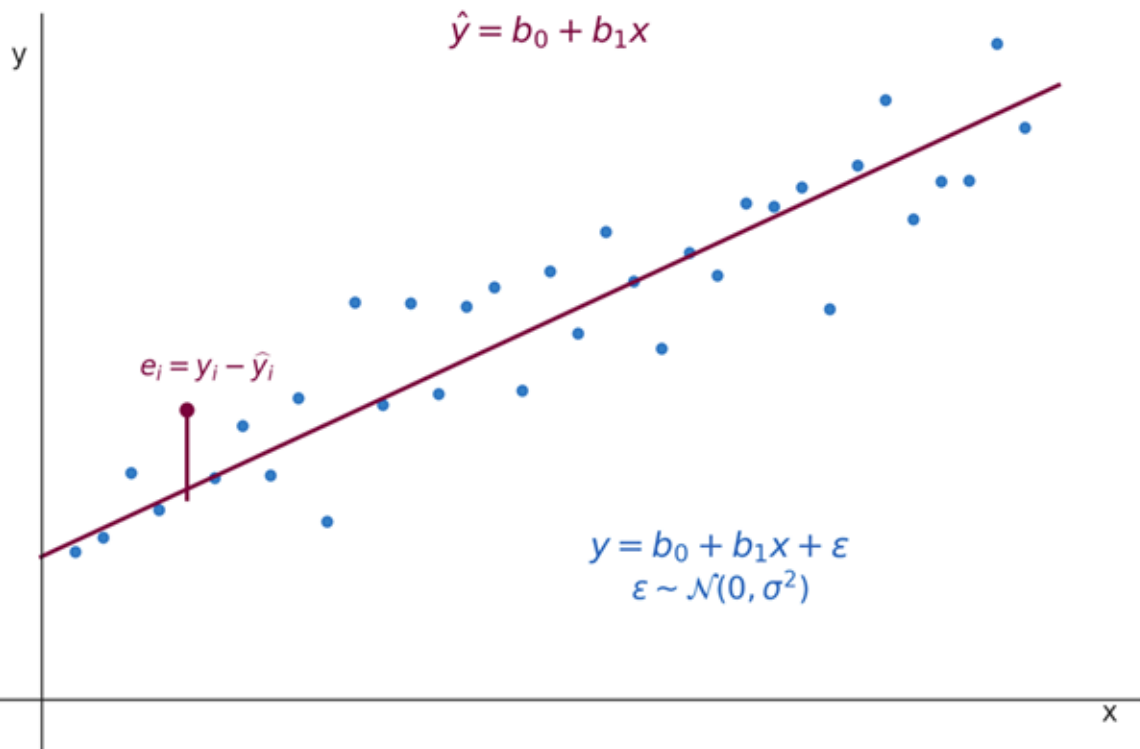
The systematic part is:

$$b_0 + b_1x_i$$

The noise term is  $\varepsilon_i$  which corresponds to the deviations from the straight line. So:

observed value = systematic part + noise

## FITTED VALUES



Once we have data, we estimate the line that best summarizes the average relationship.

$$\hat{y}_i = b_0 + b_1x_i$$

The hat,  $\hat{\phantom{y}}$  on top of the  $y$  means that it is the predicted or fitted, after smoothing over the noise.

The residual is

$$e_i = y_i - \hat{y}_i$$

so

$$y_i = \hat{y}_i + e_i$$

## NOISE VS MEASURED ERROR

The model contains unobserved noise, but after fitting the line we can measure the remaining error.

### Theoretical Model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

$$\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

### Fitted Model

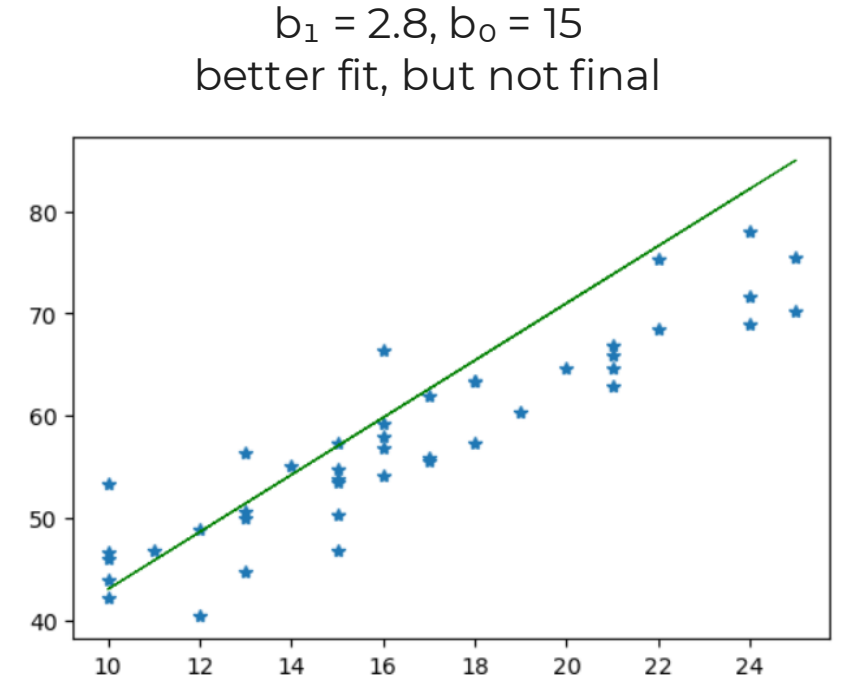
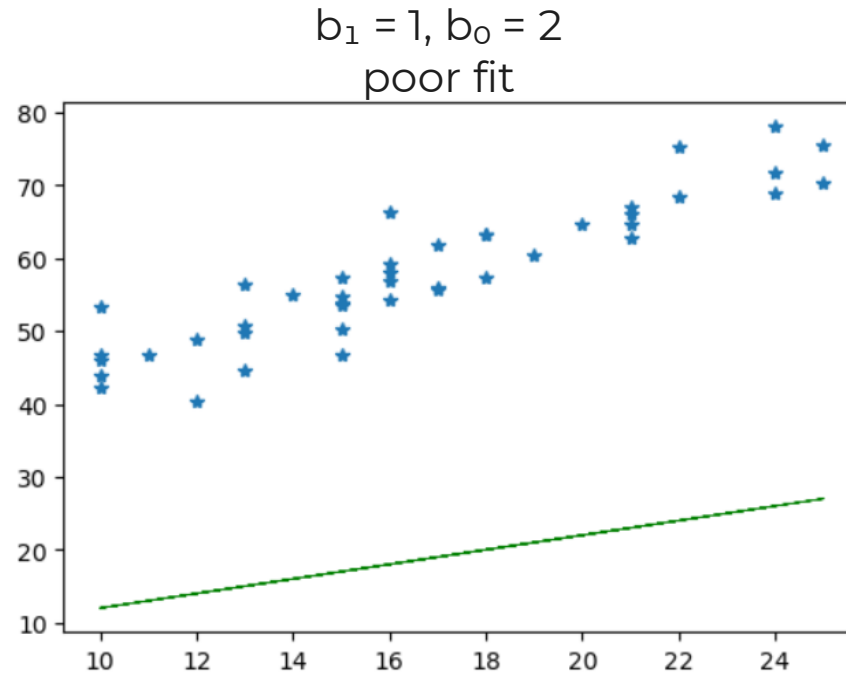
$$\hat{y}_i = b_0 + b_1 x_i$$

$$e_i = y_i - \hat{y}_i$$

$e_i \approx \varepsilon_i$  after the model is fitted.

## COMPARING TWO FITTED LINES

| Investment | Revenue |
|------------|---------|
| 10         | 46.67   |
| 13         | 44.71   |
| 21         | 65.96   |
| 17         | 55.82   |
| 16         | 57.96   |
| 22         | 75.36   |
| 15         | 53.85   |
| 15         | 46.76   |
| 13         | 56.34   |
| 15         | 50.22   |
| ...        | ...     |



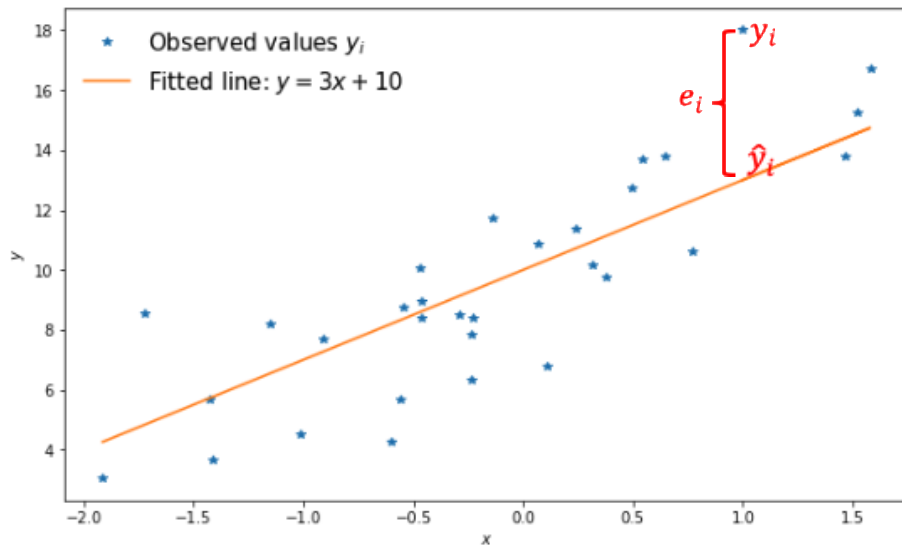
## LEAST-SQUARES FIT

We now use residuals to define the best-fitting line.

Least squares chooses the line that minimizes the **sum of squared residuals**:

$$\min \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

In other words, we choose the line that makes the **squared vertical errors** as small as possible.



### A note on “best fit”

- Least squares defines the “best” line as the one with the smallest sum of squared residuals.
- Other definitions are possible

## GOODNESS OF FIT: CORRELATION AND R-SQUARED

There are several metrics that tell how well a linear model fits the data.

**Correlation** measures the direction and strength of a linear relationship:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

with  $-1 \leq r \leq 1$



*This formula does not need to be memorized;  
correlation is computed directly in Python.*

$r > 0 \Rightarrow$  positive linear relationship

$r < 0 \Rightarrow$  negative linear relationship

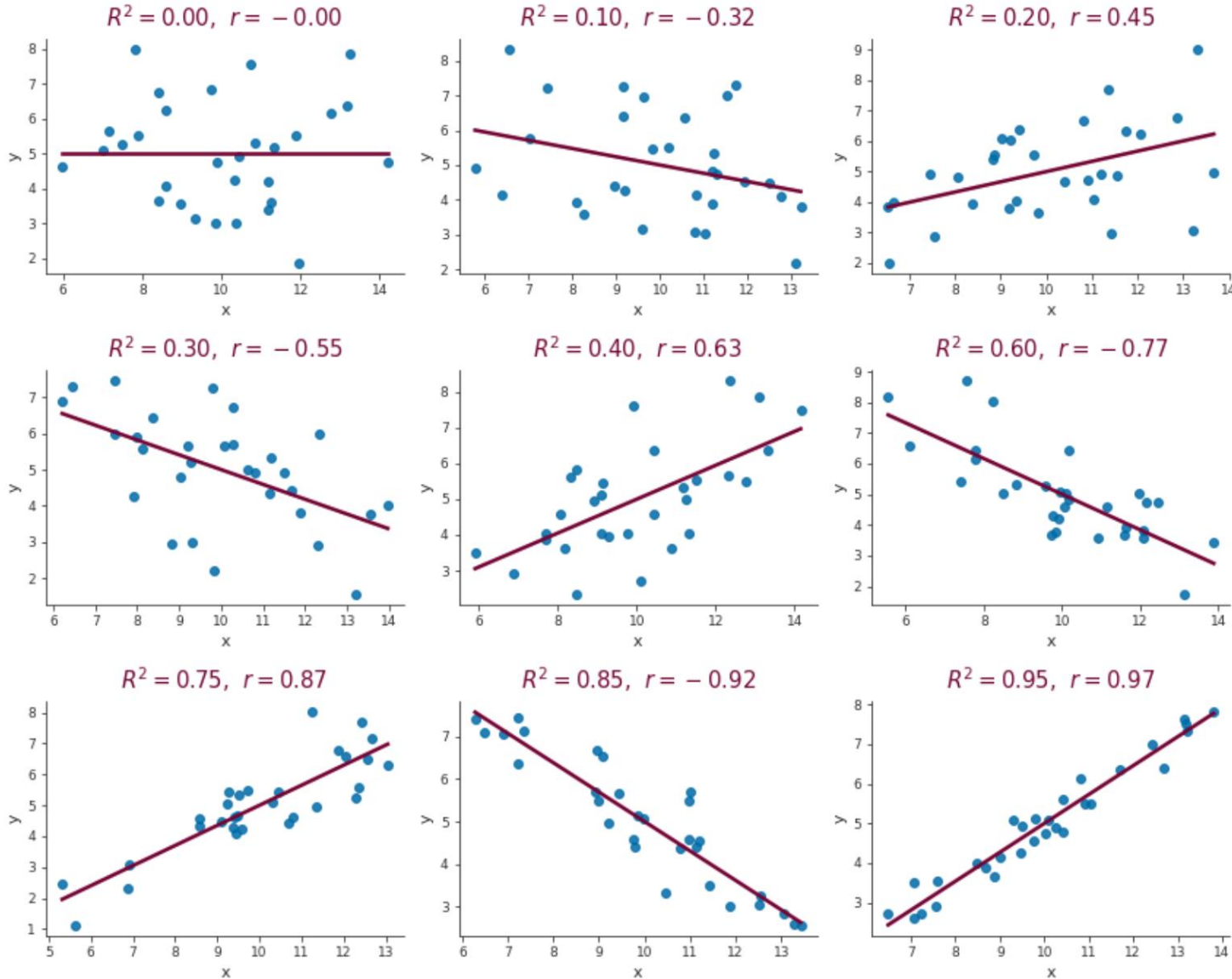
$r \approx 0 \Rightarrow$  weak or no linear relationship

$|r| \approx 1 \Rightarrow$  strong linear relationship

In simple linear regression with an intercept, a common goodness-of-fit metric is  $R^2$ , defined simply as:

$$R^2 = r^2$$

## VISUAL INTUITION FOR $r$ AND $R^2$



Correlation  $r$  keeps both direction and strength:

$$-1 \leq r \leq 1$$

The coefficient of determination  $R^2$  keeps the strength of fit only:

$$R^2 \geq 0$$

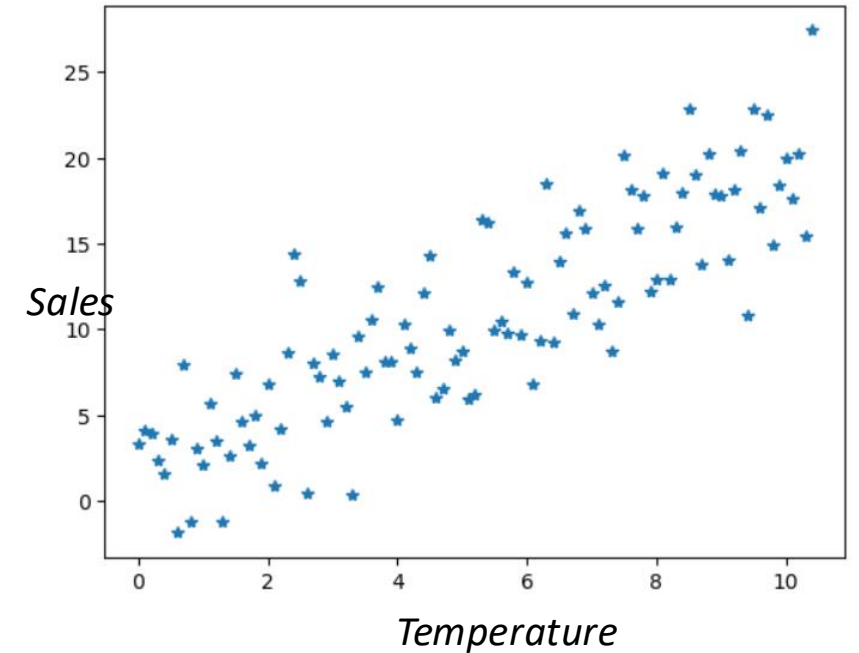
## CORRELATION VS CAUSATION

**Correlation** describes association: two variables tend to move together.

**Causation** is stronger: changing one variable directly produces a change in another.

In the scatterplot, sales and temperature move together. That association is useful, but the graph alone does not prove the causal mechanism.

Context may suggest that warmer weather affects camping sales, but regression by itself cannot prove it.



## REGRESSION IN PYTHON

```
import pandas as pd
import statsmodels.api as sm

# Read the file
df = pd.read_csv("../data/Lecture 09 teaching returns.csv")

X = sm.add_constant(df["market"])
y = df["asset_a"]

# Fit regression
model = sm.OLS(y, X).fit()

# Fitted values
df["fitted"] = model.predict(X)

# Print coefficients
print("Coefficients:")
print(model.params)

print("\nR-squared:")
print(model.rsquared)
```

```
Coefficients:
const      -0.000535
market      1.038980
dtype: float64
```

```
R-squared:
0.713885782661932
```

### Reading the code

- Load the data file.
- Assign the independent variable to X, and add a constant so that the regression includes an intercept.
- Assign the dependent variable to y.
- Create a regression model and fit it to the data.
- Use the fitted model to compute predicted values:  $y$
- The fitted model already contains the estimated coefficients and goodness-of-fit metrics.

`sm.add_constant(X)` adds a column of ones to the data so the model can estimate the intercept. Mathematically, instead of fitting only:  $\hat{y}_i = b_1 x_i$  we fit  $\hat{y}_i = b_0 \cdot 1 + b_1 x_i$  which is the same as  $\hat{y}_i = b_0 + b_1 x_i$

## WHY DO WE USE `add_constant()` ?

```
X = sm.add_constant(df["market"])
y = df["asset_a"]
```

Original  
table

| $x$        | $y$        |
|------------|------------|
| market     | asset_a    |
| 0.07362103 | 0.07966344 |
| -0.0126375 | -0.0593978 |
| 0.00731281 | 0.04192813 |
| 0.02230065 | 0.02701475 |
| -0.0255569 | -0.0077032 |

⇓

`sm.add_constant(table)`

⇓

| $X$ |            | $y$        |
|-----|------------|------------|
| 1   | $x$        |            |
| 1   | market     | asset_a    |
| 1   | 0.07362103 | 0.07966344 |
| 1   | -0.0126375 | -0.0593978 |
| 1   | 0.00731281 | 0.04192813 |
| 1   | 0.02230065 | 0.02701475 |
| 1   | -0.0255569 | -0.0077032 |

"Extended"  
table

`df["market"]` is the data structure containing our independent variable

`sm.add_constant(df["market"])` adds a column of ones so the model can estimate the intercept.

Mathematically, instead of fitting only:

$$\hat{y}_i = b_1 x_i$$

we fit

$$\hat{y}_i = b_0 \cdot 1 + b_1 x_i = (b_0, b_1) \cdot (1, x_i)$$

which is the same as

$$\hat{y}_i = b_0 + b_1 x_i$$

# INTERPRETING REGRESSION OUTPUT

Dependent variable y:  
asset\_a returns

```
print(model.summary())
```

$R^2 = 0.714$ : the model explains about 71.4% of the variation in asset\_a.

```
=====
Dep. Variable:          asset_a    R-squared:                0.714
Model:                  OLS        Adj. R-squared:           0.713
Method:                 Least Squares    F-statistic:           593.8
Date:                  Tue, 14 Jul 2026    Prob (F-statistic):    1.30e-66
Time:                  21:48:18    Log-Likelihood:        543.44
No. Observations:      240    AIC:                   -1083.
Df Residuals:          238    BIC:                   -1076.
Df Model:               1
Covariance Type:        nonrobust
=====
```

Estimated coefficients:  
 $b_0$  and  $b_1$

$b_0$  (intercept):  
predicted return when market return = 0

$b_1$  (slope / beta):  
predicted change in asset\_a when market changes by 1.

```
=====
              coef    std err          t      P>|t|      [0.025    0.975]
-----
const      -0.0005    0.002     -0.326    0.745    -0.004    0.003
market       1.0390    0.043     24.369    0.000     0.955    1.123
=====
```

95% confidence interval:  
plausible range for the true coefficient.  
For market: beta is between 0.955 and 1.123.

## MULTIPLE REGRESSION MODEL

### From one input to many

Simple regression:  $\Rightarrow \hat{y}_i = b_0 + b_1 x_i$

Multiple regression:  $\Rightarrow \hat{y}_i = b_0 + b_1 x_{1,i} + b_2 x_{2,i} + \dots + b_k x_{k,i}$

$x_{j,i}$   
= value of variable  $j$  for observation  $i$   
 $j$  = column / variable  
 $i$  = row / observation  
 $b_j$  = coefficient attached to variable  $x_j$

### Example

The process is nearly identical: `X_variables` contains all independent variables, and `X_model` adds the constant column for the intercept.

```
df = pd.read_excel("data_lecture9.xlsx",  
sheet_name="multivariate")  
  
X_variables = df[["X1", "X2", "X3", "X4"]]  
X_model = sm.add_constant(X_variables)  
y = df["Y"]  
  
model = sm.OLS(y, X_model).fit()
```

### Interpretation of the coefficients

$b_j$  = effect of  $x_j$  holding the other variables fixed

# INTERPRETING REGRESSION OUTPUT

Full regression summary:

## OLS Regression Results

```

=====
Dep. Variable:          Y      R-squared:                0.799
Model:                  OLS    Adj. R-squared:           0.789
Method:                 Least Squares    F-statistic:        76.56
Date:                  Wed, 15 Jul 2026    Prob (F-statistic):    4.66e-26
Time:                  15:33:04    Log-Likelihood:       -257.15
No. Observations:      82      AIC:                524.3
Df Residuals:          77      BIC:                536.3
Df Model:               4
Covariance Type:       nonrobust
=====
  
```

Dependent variable: Y

**R<sup>2</sup> = 0.799:**  
strong fit, but not perfect

each coefficient has its own 95% confidence interval

**b<sub>0</sub> = -4.34:**  
intercept

**b<sub>1</sub>, b<sub>2</sub>, b<sub>3</sub>, b<sub>4</sub>:**  
effects of X<sub>1</sub>, X<sub>2</sub>, X<sub>3</sub>, X<sub>4</sub> holding other variables fixed

|                | coef    | std err | t                 | P> t  | [0.025  | 0.975] |
|----------------|---------|---------|-------------------|-------|---------|--------|
| const          | -4.3359 | 4.255   | -1.019            | 0.311 | -12.810 | 4.138  |
| X1             | 2.7188  | 0.733   | 3.710             | 0.000 | 1.260   | 4.178  |
| X2             | 4.0733  | 0.246   | 16.586            | 0.000 | 3.584   | 4.562  |
| X3             | -2.2453 | 1.184   | -1.897            | 0.062 | -4.603  | 0.112  |
| X4             | 0.6958  | 0.464   | 1.500             | 0.138 | -0.228  | 1.620  |
| =====          |         |         |                   |       |         |        |
| Omnibus:       |         | 4.145   | Durbin-Watson:    |       | 1.957   |        |
| Prob(Omnibus): |         | 0.126   | Jarque-Bera (JB): |       | 2.401   |        |
| Skew:          |         | -0.183  | Prob(JB):         |       | 0.301   |        |
| Kurtosis:      |         | 2.246   | Cond. No.         |       | 65.2    |        |
| =====          |         |         |                   |       |         |        |

Now the model explains about 80% of the variation in Y.

X<sub>1</sub> and X<sub>2</sub> are clearly useful. X<sub>3</sub> has the expected negative sign but is borderline. X<sub>4</sub> is too uncertain here.

One coefficient for each input variable, plus the intercept

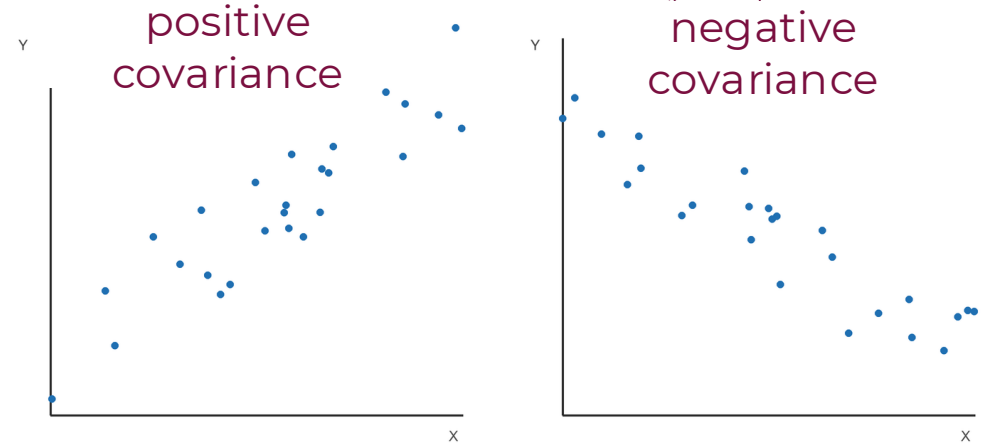
If the p-value is larger than the threshold, we do not reject the possibility that the true coefficient is zero. The estimated coefficient may be nonzero, but the confidence interval includes zero, so the evidence is not strong enough to treat the effect as clearly different from zero.

$$\hat{Y} = -4.34 + 2.72X_1 + 4.07X_2 - 2.25X_3 + 0.70X_4$$

## COVARIANCE MEASURES DIRECTION OF CO-MOVEMENT

Correlation measures linear association after standardizing units; covariance measures co-movement in the original units.

$$\text{Cov}(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$



Here,  $X$  and  $Y$  are simply two variables whose co-movement we want to measure. We are not yet assigning one as dependent and the other as independent..

Limitation: covariance depends on the units of  $X$  and  $Y$ , so its size is hard to compare across problems.

## CORRELATION STANDARDIZES COVARIANCE

Correlation is covariance divided by the standard deviations of  $X$  and  $Y$ .

$$r_{X,Y} = \frac{\text{Cov}(X, Y)}{s_X s_Y}$$

The subscript  $X, Y$  makes explicit which two variables are being compared. This is useful when the dataset contains several variables.

- q This removes units and places the result between -1 and 1.
- q Correlation measures direction and strength of linear association, but it is not a regression slope.

For simple linear regression with an intercept, this connects directly to the  $R^2$  from the previous part:

$$R^2 = r_{X,Y}^2$$

## REGRESSION SLOPE CONNECTS COVARIANCE AND CORRELATION

In simple linear regression, the slope can be written using covariance.

$$b_1 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

q The slope can be written as covariance divided by the variance of X.

$$b_1 = r_{X,Y} \frac{s_Y}{s_X}$$

q Equivalently, it is correlation multiplied by relative volatility.

| Quantity    | What it tells us                            |
|-------------|---------------------------------------------|
| Covariance  | direction of co-movement, with units        |
| Correlation | standardized strength of linear association |
| Slope       | average change in Y for one unit of X       |